Week 4, Lesson 4

Networking in AWS

Reading: Chapter 3 – Amazon Networking Services (Jaggia et al., 2020, pp. 45–71)

Outline

1. Review of computer network basics (IP, CIDR, Internet)
2. Introduction to Amazon VPC and subnets
3. Provisioning resources and connecting them in the VPC
4. Hardening AWS network security

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Review of Computer Network Basic

Internet

The Internet is a large global network of interconnected computer networks that uses the Internet Protocol Suite (TCP/IP) to facilitate communication and data exchange between devices.

- It serves as a platform for various forms of communication, including email, social media, file sharing, and web browsing.

Uses of Internet

- Communication: Provides instant messaging, video calls, and emails, enabling real-time communication worldwide.
- Information Access: Provides access to vast amounts of information via websites, databases, and online resources.
- Social Networking: Platforms like Facebook, Twitter, and Instagram allow users to connect with others and share content.
- E-commerce: Enables online shopping, banking, and financial transactions, transforming how businesses operate and consumers shop.
- Education: Offers online courses, webinars, and educational resources, making learning accessible to wider audiences.
- Entertainment: Streaming services, online gaming, and social media provide various forms of entertainment.

- The World Wide Web is one of the services that runs on the Internet.
- It enables three key technologies.

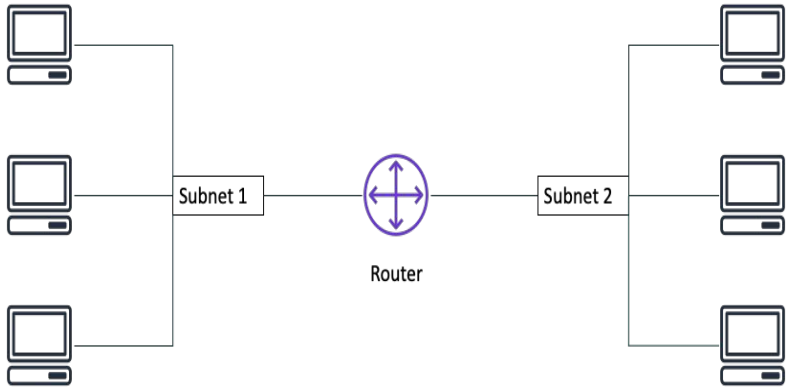
HTML (HyperText Markup Language) – for creating web pages

HTTP (HyperText Transfer Protocol) – for communication between browsers and servers

URLs (Uniform Resource Locators) – for addressing pages on the Web

Computer Network

- A computer network or data network is a telecommunications network which allows computers to exchange data.
- In computer networks, networked computing devices pass data to each other along network links (data connections).
- Data is transferred in the form of packets.
- The connections between nodes are established using either cable media or wireless media.
- The best- known computer network is the Internet.



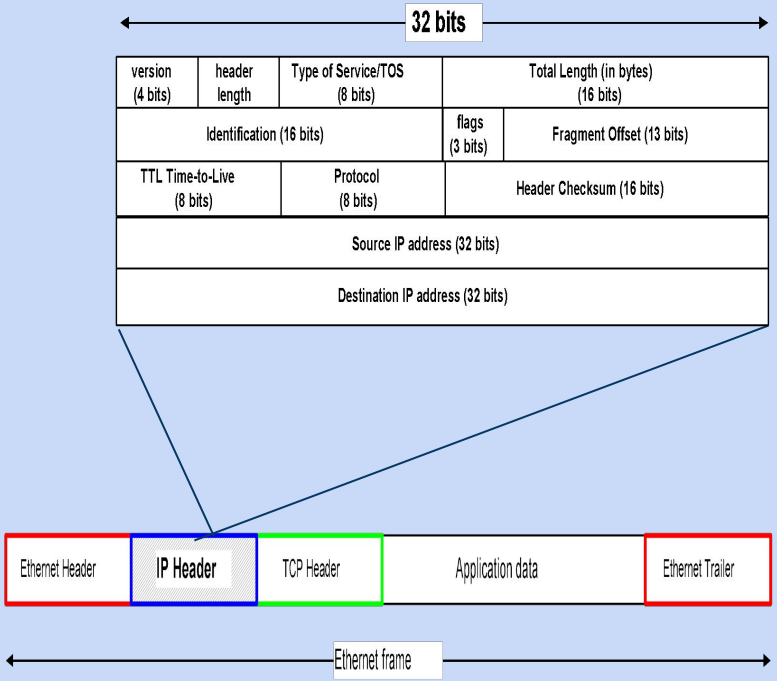
IP Addresses

- An IP Address (Internet Protocol Address) is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.

Components of an IP Address

- **Network Portion:** Identifies the network to which the device belongs.
- **Host Portion:** Identifies the individual device on the network.
- **Subnet Mask (for IPv4):** Defines which part of the IP is network and which part is host.
  - Example: IP 192.168.1.10 with subnet mask 255.255.255.0
  - Network ID: 192.168.1.0, Host ID: 10

IPv4 addressing



- Internet Protocol version 4 (IPv4) is the fourth revision in the development of the Internet Protocol (IP) and it is the first version of the protocol to be widely deployed.
- IPv4 is a connection less protocol to be used on packet-switched Link Layer networks (e.g., Ethernet).
- Version: The first header field in an IP packet is the four-bit version field. For IPv4, this has a value of 4 (hence the name IPv4).
- Internet Header Length (IHL) (4 bits) is the number of 32-bit words in the header. Since an IPv4 header may contain a variable number of options, this field specifies the size of the header .The minimum value for this field is 5 , which is a length of 5×32 = 160 bits. Being a 4-bit value, the maximum length is 15 words (15×32 bits) or 480 bits.
- Type of Services (TOS) field was for a sending host to specify a preference for how the datagram would be handled as it made its way through an internet. For instance, one host could set its IPv4 datagrams' TOS field value to prefer low delay, while another might prefer high reliability.
- Total Length: This 16-bit field defines the entire datagram size, including header and data, in bytes. The minimum-length datagram is 20 bytes (20-byte header + 0 bytes data) and the maximum is 65,535 — the maximum value of a 16-bit word. The minimum size datagram that any host is required to be able to handle is 576 bytes, but most modern hosts handle much larger packets.
- Identification: This field is an identification field and is primarily used for uniquely identifying fragments of an original IP datagram. Some experimental work has suggested using the ID field for other purposes, such as for adding packet-tracing information to datagrams in order to help trace back datagrams with spoofed source addresses.
- Flags: A three-bit field follows and is used to control or identify fragments
    - Reserved must be zero.
    - Don't Fragment (DF)
    - More Fragments (MF)
- Fragment Offset: 13 bits long and specifies the offset of a particular fragment relative to the beginning of the original unfragmented IP datagram. The first fragment has an offset of zero. This allows a maximum offset of $(2^{13} - 1) \times 8 = 65{,}528$ which would exceed the maximum IP packet length of 65,535 with the header length included.

- **Time To Live (TTL):** An eight-bit field helps prevent datagrams from persisting on an internet. This field limits a datagram lifetime. It is specified in seconds, but time intervals less than 1 second are rounded up to 1(max 255 secs).
- **Protocol**: This field defines the protocol used in the data portion of the IP datagram
- **Header Checksum**: The 16-bit checksum field is used for error-checking of the header.
- **Source address:** An IPv4 address is a group of four octets for a total of 32 bits
- **Destination address**: Identical to the source address field but indicates the receiver of the packet.
- **Options**: Additional header fields may follow the destination address field, but these are not often used

Classful addressing

Note: To calculate number of usable hosts in an IPv4 network is: 2^h-2

Class	IP Address Range	First Octet Range	Default Subnet Mask	Network Bits	Host Bits	Hosts per Network	Example IP Address
A	1.0.0.0 – 126.255.255.255	1 – 126	255.0.0.0 (/8)	8	24	16,777,214	10.0.0.1
B	128.0.0.0 – 191.255.255.255	128 – 191	255.255.0.0 (/16)	16	16	65,534	172.16.5.4
C	192.0.0.0 – 223.255.255.255	192 – 223	255.255.255.0 (/24)	24	8	254	192.168.1.10
D	224.0.0.0 – 239.255.255.255	224 – 239	N/A	N/A	N/A	N/A	224.0.0.1
E	240.0.0.0 – 255.255.255.255	240 – 255	N/A	N/A	N/A	N/A	240.0.0.5

Question 1: Given the IP address 172.16.25.30

- Identify the class of the IP address
- Write the default subnet mask
- Determine the network address
- Determine the broadcast address
- Calculate the number of hosts available

Solutions:

- Class: First octet = 172 → falls in 128–191 → Class B
- Default Subnet Mask: Class B → 255.255.0.0 (/16)
- Network Address: Keep network part, make host part = 0. This way: 172.16.0.0
- Broadcast Address: Keep network part, make host part = 255. Result: 172.16.255.255
- Number of Hosts: Host bits = 16. As from the formulla 2^16 − 2 = 65,534 hosts

IPv4 and IPv6 addresses

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Address Format	Decimal (dotted)	Hexadecimal (colon-separated)
Example	192.168.1.1	2001:0db8:85a3:0000:0000:8a2e:0370:7334
Total Addresses	~4.3 billion	~340 undecillion (very large)
Header Size	Variable (20–60 bytes)	Fixed (40 bytes)
Configuration	Manual or DHCP	Auto-configuration (SLAAC) + DHCPv6
Security	Optional (IPsec)	Built-in support for IPsec
Broadcast	Supported	Not supported (uses multicast instead)
NAT (Network Address Translation)	Commonly used	Not required
Speed & Efficiency	Slower (more overhead)	Faster & more efficient routing
Address Classes	Uses classful (A, B, C)	No classes (classless)

Classless Inter-Domain Routing (CIDR)

EIU

- CIDR (Classless Inter Domain Routing) is a method of IP address allocation and routing that allows more efficient use of IP addresses.
- Unlike traditional class-based addressing, CIDR allocates IP addresses based on a network prefix rather than a fixed class (A, B, or C)

- **CIDR Notation:** a.b.c.d/n
  - n = number of bits in the network prefix.
  - Example: 192.168.1.0/24 first 24 bits are the network, and the remaining 8 bits are the host ID.

Why CIDR needed ?

- Classful addressing wastes IP addresses:

Class	IPs Available	Hosts	Example Wastage
A	2^{24}	$2^{24} - 2$	Too large for small orgs
B	2^{16}	$2^{16} - 2$	Wastes 49,150 hosts for 214 needed
C	2^{8}	$2^{8} - 2$	Small networks only

- Problem: Organizations often need a number of hosts that do not match class sizes, leading to wastage.
- Solution: CIDR allows flexible block allocation matching exact requirements.

Rules for Forming CIDR Blocks

- All IPs must be contiguous.
- Block size must be a power of 2 (2^n) simplifies network division.
- Such as - A CIDR block is valid if: Total IPs = 2, 4, 8, 16, 32, 64… (power of 2). Starting IP is divisible by that number.
- First IP of block divisible by block size least significant bits of host ID should be 0.

For example: 192.168.10.64 → 192.168.10.79

Step 1: Calculate total IP address

79 - 64 + 1 = 16

Step 2: Find CIDR notation

16 = 2^4 → host bits = 4

32 − 4 = 28

So, CIDR = 192.168.10.64/28

Step 3: Find usable hosts

Usable = 2^h - 2

2^4 -2 = 14

Step 4: Identify addresses

Network address → 192.168.10.64

First usable → 192.168.10.65

Last usable → 192.168.10.78

Broadcast → 192.168.10.79

Step 5: Subnet mask (optional)

For /28: Total block size was 16.

Note: Block size = total IP addresses in that subnet

Therefore: 256 - 16 = 240

So, 255.255.255.240

Introduction to Amazon VPC and subnets

Cloud Networking vs Traditional Networking

Traditional Networking

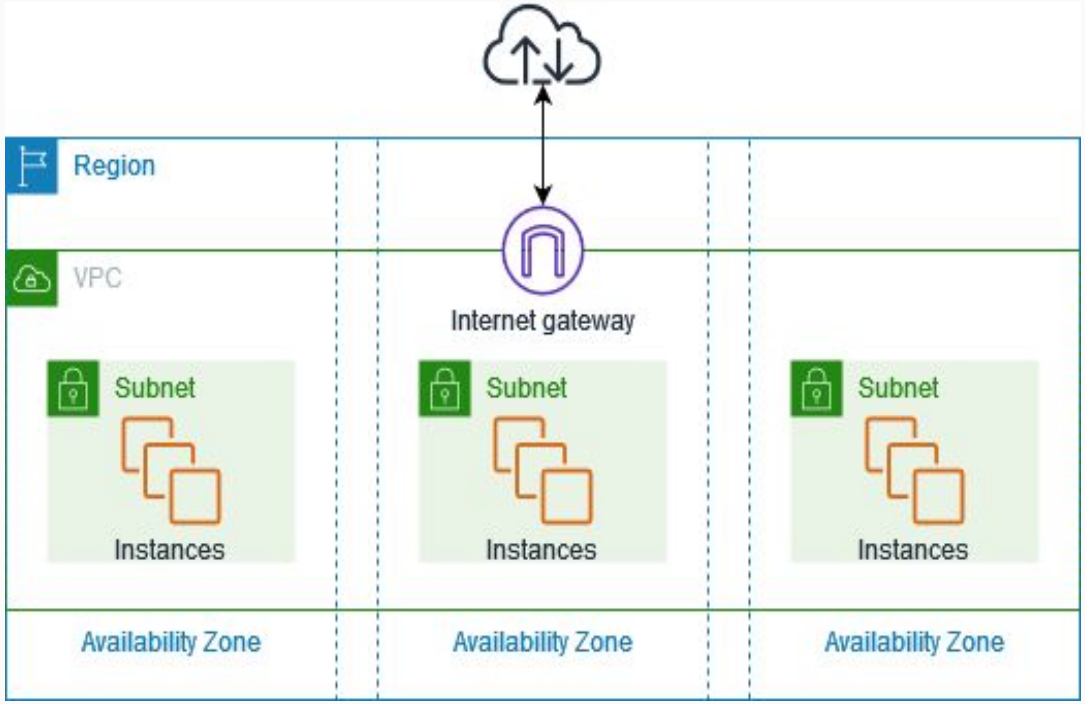
- Physical routers, switches, and cables
- Manual configuration and provisioning
- Fixed capacity and limited scalability
- Hardware replacement and maintenance
- Capital expenditure model

AWS Cloud Networking

- Software-defined virtual networks
- API-driven automation and orchestration
- Elastic scaling on demand
- AWS manages infrastructure
- Pay-as-you-go operational expenditure

AWS networking leverages Software-Defined Networking (SDN) principles to abstract physical infrastructure into programmable, flexible virtual networks. This paradigm shift enables rapid provisioning, automated configuration management, and unprecedented scalability—all without the constraints of traditional hardware limitations.

Understanding Amazon Virtual Private Cloud



- Enables you to provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define.
- Gives you control over your virtual networking resources, including:
  - Selection of IP address range
  - Creation of subnets
  - Configuration of route tables and network gateways
- Enables you to customize the network configuration for your VPC
- Enables you to use multiple layers of security

AWS VPC Architecture

- The basic architecture of a properly functioning VPC consists of many distinct services such as Gateway, Load Balancer, Subnets, etc.
- Altogether, these resources are clubbed under a VPC to create an isolated virtual environment. Along with these services, there are also security checks on multiple levels.
- It is initially divided into subnets, connected with each other via route tables along with a load balancer.

Key Components of a VPC

- **CIDR Block:** Defines the range of IP addresses for your VPC.
- **Route Tables:** Determine how traffic flows within the VPC and to external networks.
- **Internet Gateway (IGW):** Enables communication between your VPC and the internet.
- **Security Groups:** Security Groups are stateful firewalls that operate at the instance level, while NACLs operate at the subnet level.
    - Security Groups provide the primary layer of network protection for EC2, RDS, and other VPC resources.
- **Network ACLs:** NACLs are stateless firewalls that control inbound and outbound traffic at the subnet level, evaluating each packet independently.
- **Network Address Translation** (NAT): NAT Gateways allow instances in private subnets to initiate outbound internet connections while blocking unsolicited inbound traffic.

Subnets

A Subnet is a segment of a VPC's IP address range that resides entirely within a single Availability Zone (AZ).
Subnets serve two primary purposes:

- **Segmentation**: They divide large networks into smaller, isolated zones for better organization and control.
- **High Availability**: Distributing subnets across multiple AZs ensures fault tolerance and operational continuity.

Subnets are typically categorized as:

- **Public Subnets**, which route internet-bound traffic through an Internet Gateway (IGW). Used for Web servers and Load balancers.
- **Private Subnets**, which are isolated from direct internet access and communicate externally through a NAT Gateway. It is commonly used for Databases and Backend services.

Analogy

VPC as a House

- **VPC = The house itself**
  - You own it and control everything inside who enters, how rooms are arranged, and how things connect.
- **Subnets = Rooms in the house**
  - Living room (public subnet → guests can enter).
  - Bedroom (private subnet → restricted access)
- **Internet Gateway = Front door:** Allows people (internet traffic) to come in and go out
- **Route Table = Directions inside the house:** Tells visitors how to move around (which door leads where)
- **Security Groups = Door locks:** Decide who can enter each room

Summary

- VPC = your private space (house)
- Subnets = sections inside (rooms)
- You control access, layout, and security

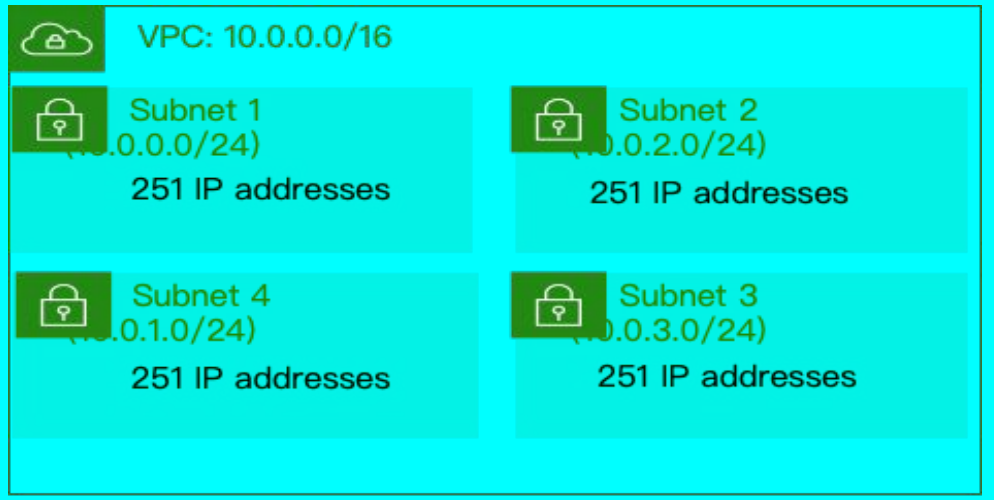
AWS IP addressing

- When you create a VPC, you assign it to an IPv4 CIDR block (range of private IPv4 addresses).
- You cannot change the address range after you create the VPC.
- The largest IPv4 CIDR block size is /16.
- The smallest IPv4 CIDR block size is /28.
- IPv6 is also supported (with a different block size limit).
- CIDR blocks of subnets cannot overlap.

- Example: A VPC with an IPv4 CIDR block of 10.0.0.0/16 has 65,536 total IP addresses.
- The VPC has four equal-sized subnets. Only 251 IP addresses are available for use by each subnet.

Public IPv4 address

- Manually assigned through an Elastic IP address.
- Automatically assigned through the auto-assign public IP address settings at the subnet level



Elastic IP address

- Associated with an AWS account
- Can be allocated and remapped anytime
- Additional costs might apply

How all works together

- VPC defines the IP range (CIDR)
- Subnets split the range
- Instances get: Private IP (always), Public IP (optional)
- Elastic IP gives a fixed public address

Route tables and routes

- A route table contains a set of rules (or routes) that you can configure to direct network traffic from your subnet.
- Each route specifies a destination and a target.
- By default, every route table contains a local route for communication within the VPC.
- Each subnet must be associated with a route table (at most one).

Main (Default) Route Table

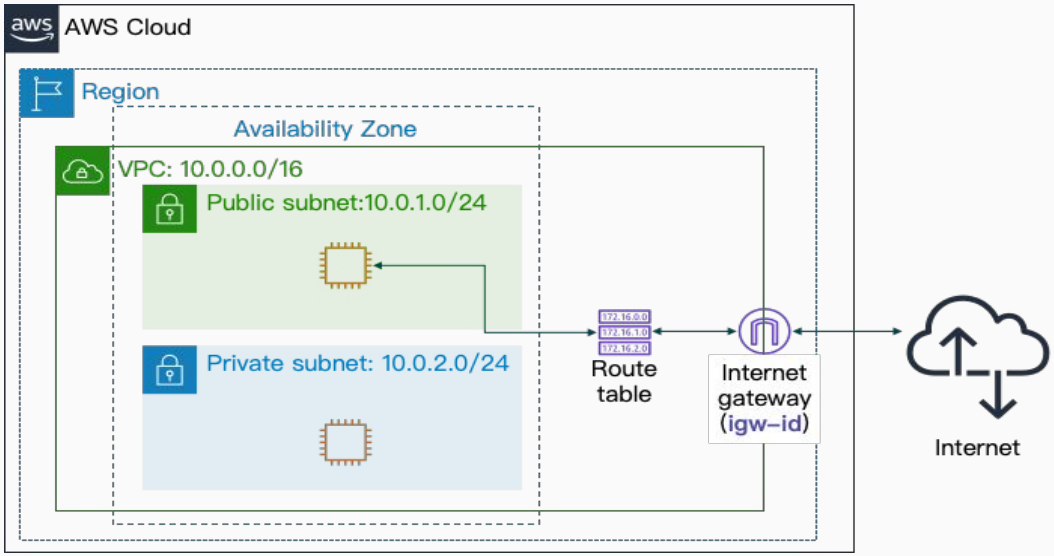
Destination	Target
10.0.0.0/16	local

VPC CIDR block

Public Subnet Route Table

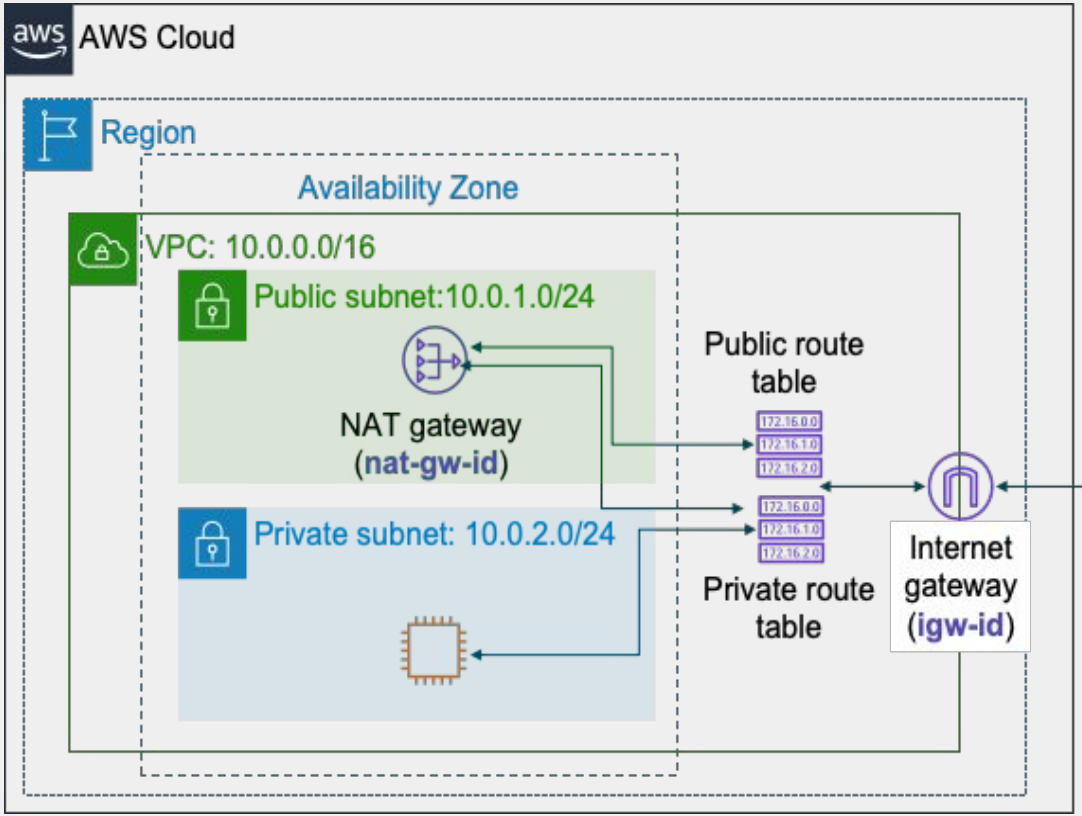
Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw–id

An **internet gateway** is a scalable, redundant, and highly available VPC component that allows communication between instances in your VPC and the internet.



Network Address Translation (NAT)

- NAT Gateways are typically placed in public subnets and act as controlled egress points for private subnets.



Public Subnet Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw–id

Private Subnet Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	nat–gw–id

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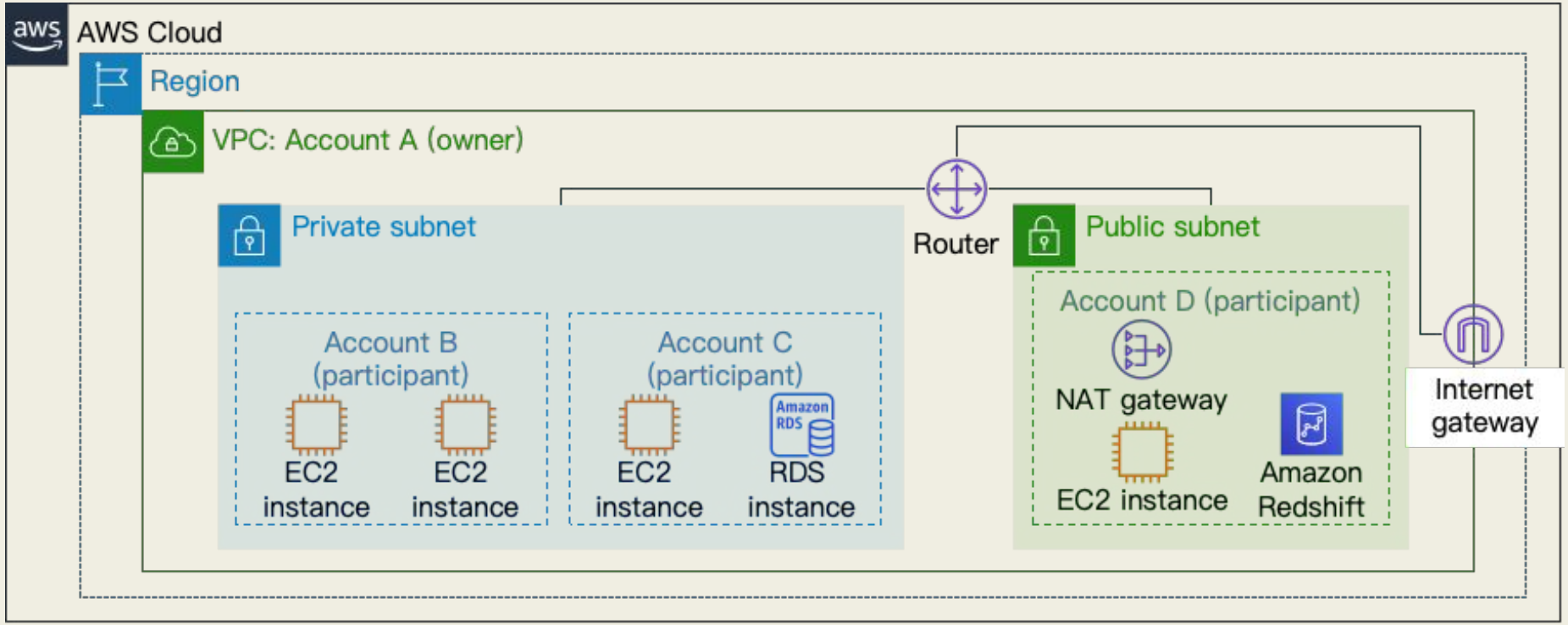
Provisioning resources and connecting them in the VPC

- **Creating a VPC and a subnet using the AWS cloud console and CloudShell:** Read chapter 3 (Amazon Networking Services (Jaggia et al., 2020, pp. 45–71)) instructions from page number 49 for creating a VPC with subnets.
  - Create VPC
  - Create subnets
  - Launch EC2 instances
  - Access the EC2 instances from the internet

Provisioning more cloud resources includes:

- VPC sharing: VPC sharing enables customers to share subnets with other AWS accounts in the same organization in AWS Organizations.
- VPC sharing enables multiple AWS accounts to create their application resources such as Amazon EC2 instances, Amazon Relational Database Service (Amazon RDS) databases, Amazon Redshift clusters, and AWS Lambda functions into shared, centrally managed VPCs.

- In this model, the account that owns the VPC (owner) shares one or more subnets with other accounts (participants) that belong to the same organization in AWS Organizations.
- After a subnet is shared, the participants can view, create, modify, and delete their application resources in the subnets that are shared with them.
- Participants cannot view, modify, or delete resources that belong to other participants or the VPC owner.

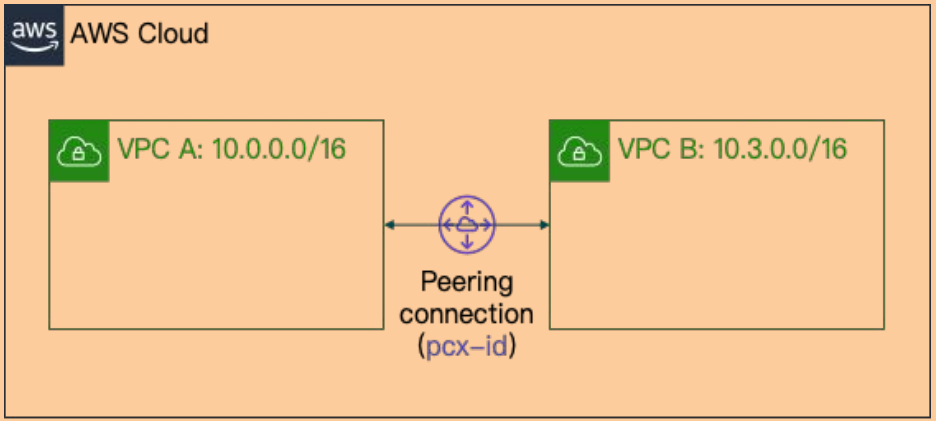


VPC sharing offers several advantages:

- Separation of duties – Centrally controlled VPC structure, routing, IP address allocation
- Ownership – Application owners continue to own resources, accounts, and security groups
- Security groups – VPC sharing participants can reference the security group IDs of each other
- Efficiencies – Higher density in subnets, efficient use of VPNs and AWS Direct Connect
- Optimized costs – Costs can be optimized through the reuse of NAT gateways.

VPC peering

- A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately.
- Instances in either VPC can communicate with each other as if they are within the same network.
- You can create a VPC peering connection between your own VPCs, with a VPC in another AWS account, or with a VPC in a different AWS Region.



Route Table for VPC A

Destination	Target
10.0.0.0/16	local
10.3.0.0/16	pcx-id

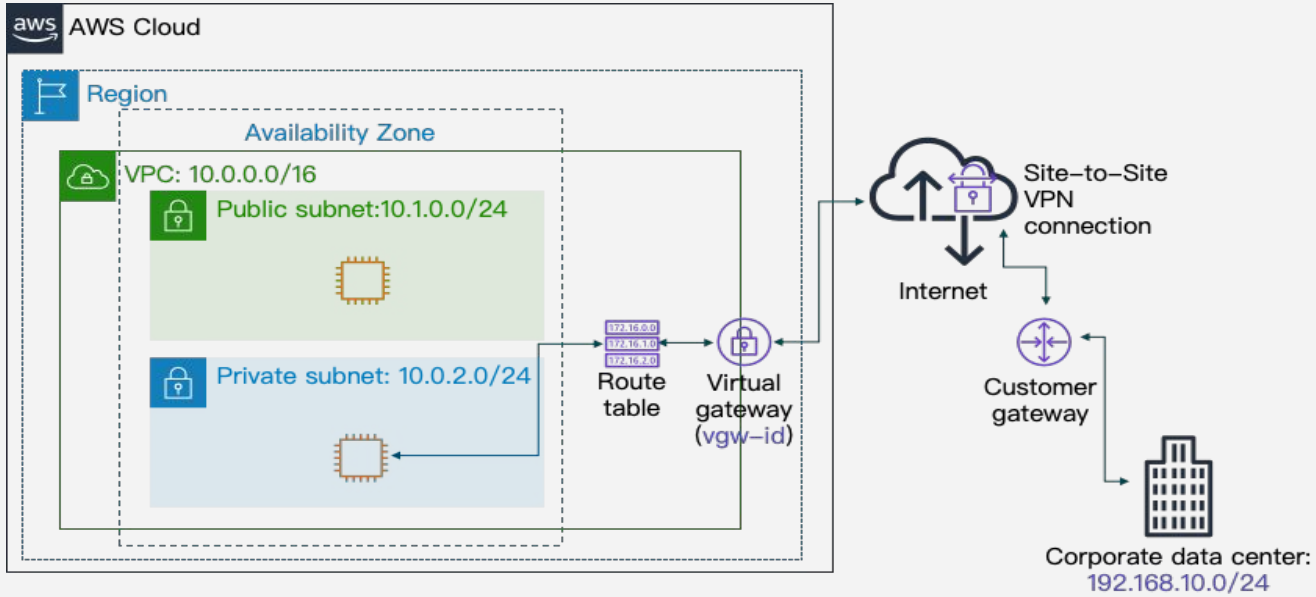
Route Table for VPC B

Destination	Target
10.3.0.0/16	local
10.0.0.0/16	pcx-id

Restrictions:

- IP spaces cannot overlap.
- Transitive peering is not supported.
- You can only have one peering resource between the same two VPCs.

AWS Site-to-Site VPN: By default, instances that is launch into a VPC cannot communicate with a remote network.



- To connect the VPC to remote network (that is, create a virtual private network or VPN connection.
- Create a new virtual gateway device (called a virtual private network (VPN) gateway) and attach it to your VPC.
- Define the configuration of the VPN device or the customer gateway.
- The customer gateway is not a device but an AWS resource that provides information to AWS about your VPN device.
- Create a custom route table to point corporate data center-bound traffic to the VPN gateway. Update security group rules.
- Establish an AWS Site-to-Site VPN (Site-to-Site VPN) connection to link the two systems together.
- Configure routing to pass traffic through the connection.

Public subnet route table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw–id

Private subnet route table

Destination	Target
10.0.0.0/16	local
192.168.10.0/24	vgw–id

Simple analogy

- Creating VPC + subnets = building a house
- Provisioning resources = putting furniture inside
- Connecting = wiring electricity, doors, and pathways

Hardening AWS network security

Hardening AWS network security means strengthening the protection of your VPC and the resources inside it so they are safe from unauthorized access, attacks, or misuse.

Why security need ?
Once your VPC resources (like EC2 instances) are connected to the internet:
"security has become a very important factor"
So, need to controls Limit access, Prevent attacks, and Protect data.

VPC firewalls

- Security Groups act like a virtual firewall for EC2 instances.
- It control inbound and outbound traffic
- SG is stateful
- Allow/deny based on:
  - IP address
  - Port number (e.g., 22 for SSH)
  - Protocol (TCP, UDP)
- So, if inbound traffic is allowed, the response is automatically allowed back.
- **For example**: Security Groups = Door Locks on Each Apartment
  - Each apartment (EC2 instance) has its own lock
  - Only allowed people can enter
  - Only people with a key can open the door (only specific ports/IPs allowed)

Network ACLs (Subnet-level firewall)

Works for all instances in a subnet

- Allows or denies traffic → Based on IP, port, and protocol
- Checked before Security Groups → First level of filtering
- Uses numbered rules → Smaller number = higher priority
- NACL is Stateless → Must allow both:
  - Incoming traffic
  - Outgoing response

Feature	Security Group	NACL
Level	Instance	Subnet
State	Stateful	Stateless
Rules	Allow only	Allow & Deny
Applies to	Specific instances	Entire subnet

Example:

- Network ACLs = Security Guard at the Building Entrance
- Guard checks everyone entering AND leaving
- Can allow or deny people
- Important difference:
  - Guard checks both directions every time (stateless)

Principle of Least Privilege

  - Only allow necessary traffic
  - Reduce attack surface

Example:

- Don’t open all ports (0–65535)
- Only allow required ports like:
- 22 (SSH)
- 80 (HTTP)

VPC endpoints

Instead of using the internet:

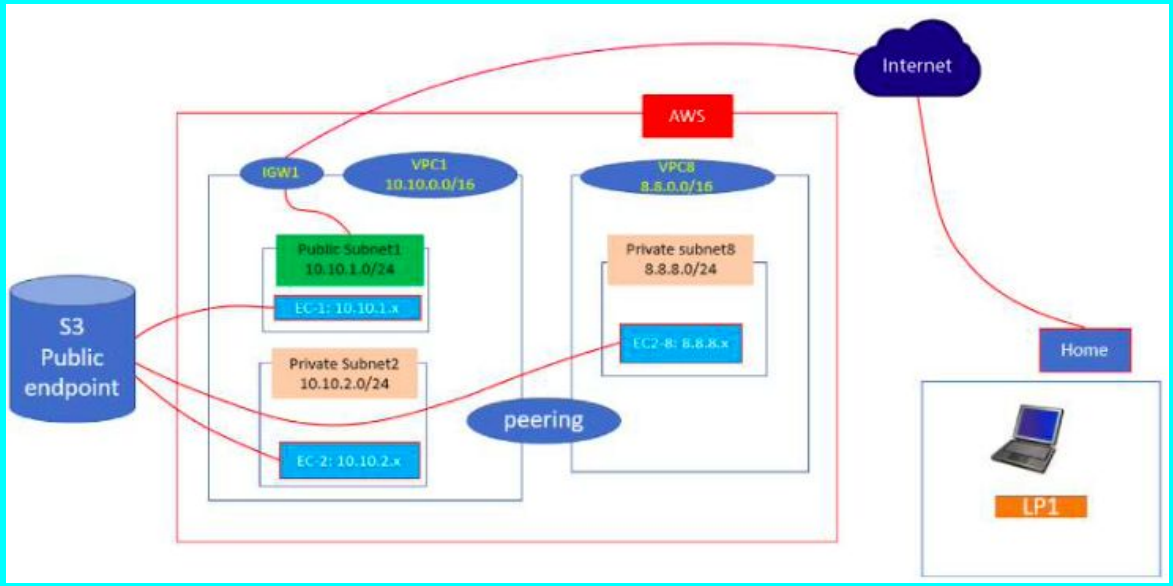
- Use VPC Endpoints to access services like S3 privately

Advantages:

- No internet exposure
- More secure communication

Example:

- EC2 in private subnet can access S3 without Internet Gateway or NAT



Component	Purpose	Type
Security Group	Control traffic to EC2	Firewall
NACL	Control traffic in subnet	Firewall
VPC Endpoint	Private access to services	Connectivity

Create a VPC endpoint for S3 in the us-east1 region:

- Log in to the AWS Management Console and navigate to the Amazon VPC console. Click on **Endpoints** in the left-hand menu. Click the **Create Endpoint** button.

- Fill in the endpoint settings in the new pop-up window.
    - vpc-s3-endpoint is the name
    - Select **AWS services** as the service category
    - Select **com.amazonaws.us-east-1.s3** as the service name
    - Select **VPC1** to create the endpoint
    - Select the **VPC1-Pri** route table to associate with the endpoint
    - Enable policies for resource-based policies to control access to the bucket
    - Click **Create endpoint** to create the VPC endpoint.
- Now you can verify S3 access from EC2-2, which does not have internet access.

School Building Analogy

- Imagine your **VPC** is a **school building**.
- When you first build the school, anyone might walk in that's **not secure**. So you "harden" security by adding protection.

1. **Security Groups = Classroom Door Locks**
    - Each classroom (EC2 instance) has its own lock
    - Only certain students/teachers can enter
    - Example: Only teacher can enter staff room (port 22 allowed)
2. **Network ACLs = School Main Gate Security**
    - Guards at the school entrance
    - They check everyone entering and leaving
    - Important: You must allow both entry AND exit. Otherwise students may get stuck
3. **Least Privilege = Only Necessary Access**
    - Not every student can enter:
        - Principal office
        - Staff room
        - Only give access where needed
4. **VPC Endpoint = Secret Internal Path**
    - Students go to library using internal corridor
    - No need to go outside the school (internet)

Further reading

- Chapter 3 – Amazon Networking Services (Jaggia et al., 2020, pp. 45–71)
- https://docs.aws.amazon.com/vpc/latest/userguide/what-is-amazon-vpc.html
- AWS Site-to-Site VPN and other VPN connectivity options, see VPN Connections in the AWS Documentation at https://docs.aws.amazon.com/vpc/latest/userguide/vpn-connections.html.
- VPC peering, see VPC Peering in the AWS Documentation at https://docs.aws.amazon.com/vpc/latest/userguide/vpc-peering.html.
- NAT gateways at https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html.
- NAT instances at https://docs.aws.amazon.com/vpc/latest/userguide/VPC_NAT_Instance.html.
- Differences between NAT gateways and NAT instances at https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html.
- Internet Gateways in the AWS Documentation at https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Internet_Gateway.html.

Thank you!

End